Meet Babariya

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Summary

International Business Information Systems student at Furtwangen University (grade **1.55**) and ML Benchmark Engineer at **Datacurve (Y Combinator)**. **Kaggle Competitions Expert** (top 1%, silver medal), **12th in Germany** at IMC Prosperity 4 and WorldQuant BRAIN Gold. Experienced in Python, PyTorch, LLM agents and evaluation, and in taking ML systems from data to deployment. Seeking an **AI / ML engineering internship** in Germany.

Experience

Datacurve (Y Combinator) — Project Eris Remote
ML Benchmark Engineer (Contract) *Sep 2026 – Present*

- Design end-to-end ML benchmark challenges used to train and evaluate AI coding agents: dataset curation, split pipeline (*prepare.py*), evaluator (*grade.py*), scoring rubrics and reference solution notebooks.
- Built 25+ challenge designs across **vision fine-tuning, sequence-to-sequence, from-scratch and LLM-evaluation** tracks, on licensed real-world data such as **multi-channel sensor and telemetry records** and scientific images.
- Calibrated difficulty so frontier agents score below 0.8 (baseline ladders up to GPU fine-tunes, exact Bayes-optimal decoders to prove information ceilings) and hardened graders against label leakage and test-set lookup.

Jainam Broking Surat, India
Derivatives Trader *Apr 2022 – Dec 2024*

- Traded weekly NIFTY / BANKNIFTY options with delta-neutral strategies and real-time, Greeks-based hedging.

Education

Furtwangen University (HFU) Furtwangen, Germany
B.Sc. International Business Information Systems *Mar 2025 – Present*

- Current grade: **1.55** · Transcript

Projects

EV-Motor Virtual Temperature Sensor + LLM Engineering Assistant | Case study & live demo | GitHub
Python, PyTorch, scikit-learn, Gemini API, Vercel, Streamlit

- Estimates rotor-magnet and stator temperatures of a PMSM from vehicle-measurable signals (185 h test-bench data). Ensemble of a load-history linear model and a physics-structured **thermal neural network**: **1.50 K² MSE / 5.73 K worst case** on the published benchmark (best published: 2.87 / 6.02), without torque as an input.
- Found that the benchmark's test cycles contain only cold starts. After a hot restart a cold-start assumption is ~30 K wrong for 40 min, which I fixed with stored-state initialisation. All choices were made by grouped cross-validation.
- LLM assistant (hand-written agent loop, 6 deterministic tools, fallback across free models) deployed as a **serverless API**. It scores **13/13** on a graded evaluation in production; the evaluation caught 3 real bugs.

QuantPilot — AI copilot & quant-ranked copy trading | *TypeScript, Next.js, Python, FastAPI, PostgreSQL* *code private*

- Full-stack product on Hyperliquid: TypeScript execution API and bot over a Python service that ranks traders by risk-adjusted returns with a Deflated-Sharpe filter (run on live mainnet data).

Kaggle Silver Medal — CIBMTR: Equity in post-HCT Survival Predictions | Write-up | *GBDTs, PyTorch (TabM), AutoGluon*

- **120th / 3,325 teams** (team of three). Predicted event-free survival after stem-cell transplants, scored by a race-stratified concordance index so the model must be accurate *and* fair across patient groups.
- Turned censored survival data into regression targets (Kaplan–Meier, Nelson–Aalen) and blended gradient-boosted trees with tabular neural networks in a two-level stack, using 10-fold CV. Private leaderboard: 0.693.

Competitions & Achievements

- **Kaggle Competitions Expert** (top 1% globally): Silver, CIBMTR (see above); Bronze, MITSUI Commodity Prediction (165th / 1,711).
- **IMC Prosperity 4**: 230th / 18,803 teams, **12th in Germany** · **WorldQuant BRAIN** Gold · AlgoTrade Hackathon 2025 · WQU Applied Data Science Lab.

Technical Skills

LLM / GenAI	Claude, Gemini & OpenAI APIs, agents & tool use, RAG, embeddings, vector DBs (ChromaDB), LLM evaluation, benchmark & rubric design
ML / Deep Learning	PyTorch, scikit-learn, time-series feature engineering, physics-structured / state-space models, grouped cross-validation, Hugging Face, CNNs
Software	Python, JavaScript / TypeScript, SQL; FastAPI, Streamlit, Vercel serverless, Next.js / Astro, PostgreSQL, Docker, Git, GitHub Actions; AI-assisted coding (Claude Code); pandas, NumPy, Power BI
Office	MS Office: Excel (incl. VBA), PowerPoint, Word, Teams, Outlook
Languages	English (fluent), German (A2, actively learning)